WACC Calculation

Gross vs net debt, cost of debt, WACC, hybrids & preferred stock

Gross vs Net	Cost of debt	Synthetic rating	Subsidised debt	WACC	Hybrids

CARRIED OVER FROM MODULE 2

Remaining concerns when using foreign comparables (Brazilian aerospace example): ***C. Operating leverage, D. Government exposure, E. Capital structure.***

1. Gross Debt vs Net Debt

	Gross D/E	Net D/E
Formula	*Debt / Equity*	*(Debt − Cash) / Equity*
Idea	*Looks at all borrowings*	*Cash reduces net financial exposure*

EXAMPLE

Suppose you have ***₹100 of business assets*** *financed with* ***₹20 debt + ₹80 equity****. Now imagine you also hold* ***excess cash of ₹20****. Having excess cash* ***effectively reduces the company's net financial exposure****.*

NOTE

You ***cannot mix a net debt beta with a gross debt ratio*** *elsewhere in your valuation.* ***Be consistent with net debt beta.***

Where net debt must be used consistently

#	Where	What to do
1	*Levered beta*	$\beta_L = \beta_u \times [\ 1 + (1 - t) \times$ **Net D/E** $\ ]$, *where Net D = Debt − Cash*
2	*WACC*	$w_E = E / (E + \text{Net D})$ *and* $w_D = \text{Net D} / (E + \text{Net D})$. K_d *is the cost of* ***actual, not net, debt****.*
3	*Enterprise value*	*EV = Equity value + Debt − Excess cash. Same for both gross or net debt beta.*
4	*Bottom-up beta*	*Unlever / relever using* ***net D/E*** *while calculating bottom-up beta.*
5	*Capital structure assumptions*	*When forecasting the future capital structure, stick with gross D/E* ***or*** *net D/E.*

V.V. IMP: CASH ON THE BALANCE SHEET ≠ EXCESS CASH

Excess cash is over and above the **minimum operating cash balance**. Remove cash required for operations, investments (capex), acquisitions and commitments, present or future.

To find minimum operating cash: a) look at historical balances; b) use the cash flow cycle (FM); c) **management disclosures** (best practical evidence).

Prefer the Gross Debt approach.

2. Cost of Debt

NOTE

Do **not use historical K_d** as leverage and interest rates change with time. We want the **current** cost of debt.

Two ways to estimate current cost of debt

- **a) Look at an actual traded bond.**
- **b) Use the company's credit rating.** Say the company is A-rated: look at the default spread associated with A-.

$$K_d = R_f + \text{default spread}$$

EXAMPLE (US COMPANY)

R_f = 4.29%, A- spread = 1%. So K_d = **4.29% + 1% = 5.29%**.

Multiple ratings? Use a **representative median rating**: take some comparable companies, sort their ratings, then take the median.

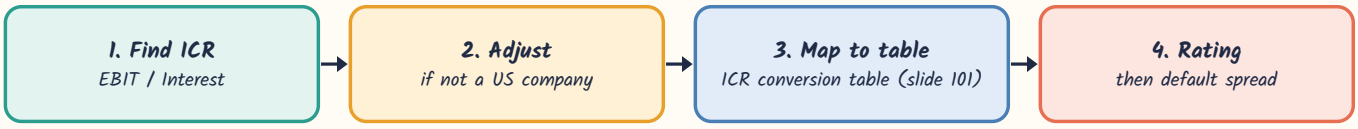
No credit rating? Create a synthetic rating

Key variable: **Interest Coverage Ratio**.

$$\text{ICR} = \text{EBIT} / \text{Interest}$$

LOGIC

Higher ICR ⇔ lower default risk ⇔ higher credit rating ⇒ lower default spread.



Country-level cost of debt: two methods (prefer the 1st)

	Method 1 (preferred)	Method 2 (alternative)
Formula	$K_d = R_f^{country}$ + company credit spread	$K_{d,country} = R_f^{USD}$ + country spread, then convert currency (below)
Spread	Company spread = Indian corporate YTM − Indian govt YTM	$(1 + K_{d,country}) = (1 + K_{d,USD}) \times (1 + \pi_{country}) / (1 + \pi_{USD})$, π = inflation

In the 1st method: **$R_f^{country} = R_f^{USD}$ + country default spread**. Do **not** add the whole country default spread, only the **exposure to the country's risk**: a company may have ⅓ of its revenue outside the country.

EXAMPLE: EMBRAER

R_f^{USD} = 4.29%; country default spread = 6.01%; company default spread = 1% (derived from A rating). ⅓ of Embraer's revenue is from the US, so CRP on that part = 0%.

K_d = 4.29% + ⅔ (6.01%) + (0% × ⅓) + 1% ≈ 9.29%

Synthetic rating caveats

The ICR table was built using **US companies**. We cannot blindly apply it to other countries, because the ICR-to-rating relationship depends on the **overall corporate lending rate** in the country.

	USA (rate 5%)	India (rate 10%)
Interest	Debt × 5%	Debt × 10%
ICR	EBIT / (Debt × 5%)	EBIT / (Debt × 10%)
Same EBIT & debt	12x	6x (ICR halves)

So if emerging markets carry **twice the interest burden** for the same underlying business risk, **double the ICR before using the US table**. E.g. Indian company ICR = 6x → convert (×2) → 12x.

Adjusted ICR = $ICR_{country\ company}$ × (Country corporate rate / US corporate rate)

Default spreads change over time

- Investors' perception of credit risk changes. **In a recession investors demand more compensation** for holding risky bonds, so default spread ↑. The reverse happens when markets are very confident.

- **Lower rating ⇔ higher default spread.**

KEY IDEA

A BBB company in a financial crisis may have a **much larger spread** than a BBB company in a booming economy. The rating has not necessarily changed, but **the price of credit risk has**.

For valuation you want the **current cost of borrowing**, not some historical number.

3. Subsidised Debt

Suppose **government-supported debt costs 6%** while comparable **market debt costs 9%**. Distinguish the **actual contractual cost** from the **market-required return** for the underlying credit risk.

- A financing subsidy is an **economic benefit**.
- If the subsidised rate is used in WACC, **treat the subsidy consistently**.

Prefer the market cost, i.e. 9%

4. WACC

$$WACC = K_e \times E/(D+E) + K_d \times (1 - T) \times D/(D+E)$$

- **Always use market value weights.**
- For **unlisted companies** there is no market value, so use the **comparable-company approach**: take the weights from comparable companies.

5. Hybrids & Preferred Stock

PART 1: CONVERTIBLE BOND

Convertible bond = straight debt + conversion option. The conversion option behaves like equity. **Separate the two.**

PART 2: PREFERRED STOCK

Keep it as a **separate component** in the capital structure with its own cost:

$$K_p = \text{Preferred dividend } (P_D) \,/\, \text{Market price (M.P)}$$

Example: splitting a convertible bond

Face value	Coupon	Maturity	Market value	Rating	Equivalent A-rated straight bond
$125 million	4%	10 years	$140 million	A	8%

To find the debt position: interest = 4% × $125m = **$5m p.a.** An equivalent A-rated bond requires an 8% return, so discount at 8%:

$$\text{Debt value} = 5 \times PVIFA_{8\%,10} + 125 \,/\, (1.08)^{10} \approx \$91.45\text{m}$$

Convertible bond = $140m

Debt $91.45m: add to debt	Option $48.55m: add to equity

Conversion option = 140 − 91.45 = **$48.55m**, added to equity.